

NDC-01 Mission-Critical Data Center Cooling Modernization

*Engineering Project Management + Systems Engineering + Reproducible Evidence
Engineering Evidence Integrity Release Candidate v6.1*

Independent simulated case study. All organizations, assets, operating data, costs and results are fictional.

1. Executive scenario

NovaTech Data Centers (NDC) is a fictional mid-sized colocation and digital-infrastructure provider. NDC-01 is a regional facility serving approximately 80 enterprise customers, with 420 installed racks, approximately 350 occupied racks and a modeled critical IT load up to 800 kW. The facility operates 24x7x365.

The cooling plant comprises three 400 kW chillers (1.2 MW nominal capacity), six pumps, field instrumentation, PLC-based control and a legacy BMS/SCADA layer. Under the defined design condition, two chillers support the critical cooling requirement while the third provides intended redundancy.

2. What was happening before the project

During the 2026 baseline period, Facilities Engineering identified progressive deterioration rather than one catastrophic outage. Repeated equipment events were consuming maintenance capacity, reducing redundancy margin, increasing thermal exposure and worsening operating economics. Equipment downtime is explicitly not treated as IT-service downtime.

Baseline evidence	Value
Equipment failures	13
Equipment downtime	121.75 h
Outside intended N+1 state	66.75 h
Cooling-capacity deficit	4.75 h
Cooling-service availability	99.9458%
PM compliance	72.22%
Weighted COP	2.809
Modeled PUE	1.551
Cooling O&M	EUR 425,432.79/year

3. Why the problem matters

A single cooling-asset failure can be absorbed by redundancy; however, the facility then operates with reduced resilience. A coincident event or high thermal demand can create a capacity deficit. The engineering problem is therefore not merely asset downtime, but the combination of reliability, available capacity, redundancy exposure, thermal margin, maintainability, energy performance and lifecycle cost.

4. Engineering assessment before project authorization

Management authorizes an engineering assessment before authorizing capital work. The assessment establishes baseline evidence, stakeholder needs, measurable requirements, alternatives, risk exposure and the business case. The project therefore begins with a problem and evidence, not with a predetermined technology purchase.

5. Stakeholder needs and Systems Engineering

Stakeholder	Need
Operations	Sufficient cooling capacity throughout required operation; visibility of redundancy and thermal margin.
Maintenance	Earlier degradation detection and restoration within maintainability limits.
Engineering	Reliable data for performance, degradation and energy analysis.
Finance	Measurable lifecycle value.
Service Management	Avoid cooling conditions that materially affect contracted IT services.
Cybersecurity	Logical separation and least privilege across control, supervisory and analytics functions.

Systems Engineering maintains the technical evidence chain: business concern -> stakeholder need -> system requirement -> architecture/interface -> verification evidence -> validation against intended use.

6. Key verifiable requirements

Requirement	Acceptance criterion
SYS-CAP-001	Cooling-service availability $\geq 99.95\%$
SYS-RED-001	Redundancy state and capacity margin visible for each 15-minute performance interval
SYS-MNT-001	PM compliance $\geq 95\%$
SYS-ENE-001	Weighted COP ≥ 3.30
SYS-PUE-001	Modeled PUE ≤ 1.50 within declared boundary
SYS-MON-001	Daily 1-minute monitoring completeness $\geq 95\%$
SYS-SEC-001	Defined separation/access-control evidence accepted

7. Alternatives and decision

Alternative	Indicative CAPEX	Decision logic
Continue as-is	EUR 0	Retains reliability, redundancy and cost exposure
Full plant replacement	~EUR 900k	High renewal potential; high CAPEX/disruption
Controls/instrumentation only	~EUR 130k	Improves visibility; limited reliability benefit
Targeted modernization	EUR 360k	Selected balance of reliability, controls, monitoring, energy and implementation risk

8. Project Management role

After the engineering alternative is selected, Project Management governs delivery: charter, scope/WBS, schedule, cost baseline, procurement, stakeholders, risks, resources, communications, change control, EVM, decision gates, transition and closure. Planned duration is 32 weeks and authorized funding is EUR 360,000.

At week 20, PV=EUR 210k, EV=EUR 195k and AC=EUR 205k, producing SPI 0.929 and CPI 0.951. A VFD supplier delay becomes an issue and is handled through CR-004, preserving assessment, approval and regression testing.

9. Integrated PM + SE execution

PM controls delivery of the change; SE controls technical integrity of the solution. Verification activities must appear in the schedule and budget; procurement must preserve technical requirements; changes must be evaluated for requirement/interface impact; and acceptance must be supported by evidence.

10. v6 physical and data model

The v6.1 model explicitly links time-varying IT load to cooling demand, available chiller capacity and redundancy state. It separates asset reliability from cooling-service availability and from any downstream IT-service consequence.

15-minute performance evidence contains IT load, cooling demand, cooling electrical energy, COP, modeled facility energy and PUE. A separate daily completeness file represents expected versus observed one-minute monitoring samples. Independent random-number streams reduce accidental coupling between load, failures, repair, maintenance, alarms, cost and data imperfections.

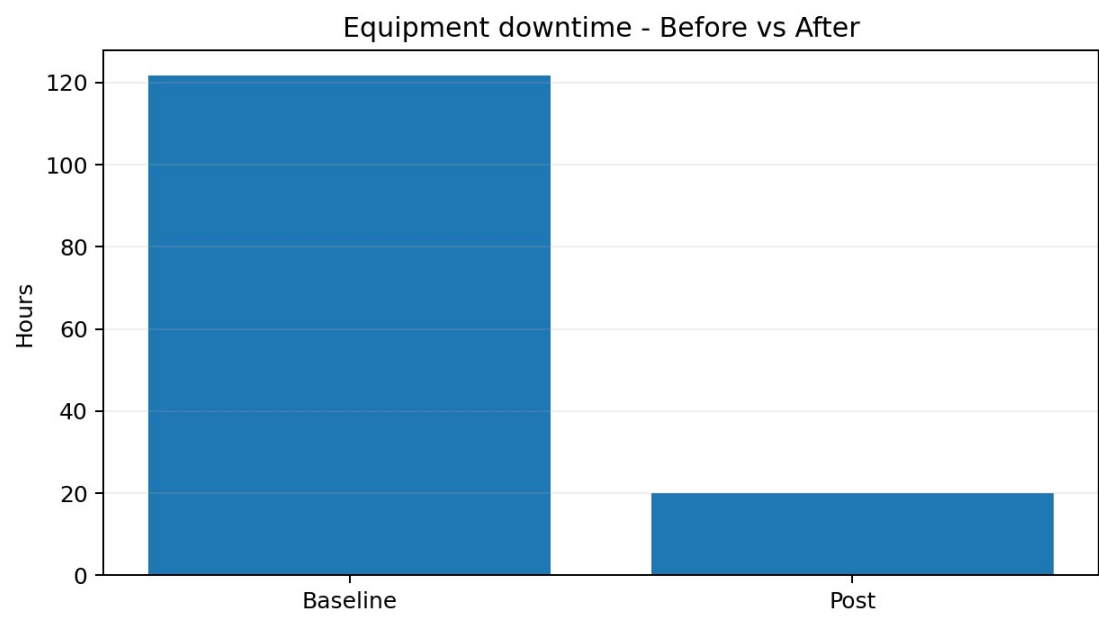
11. Post-modernization evidence

KPI	Baseline	Post
Failures	13	5
Equipment downtime	121.75 h	20.08 h
Cooling-service availability	99.9458%	99.9943%
Outside intended N+1	66.75 h	15.25 h
Capacity deficit	4.75 h	0.50 h
Weighted COP	2.809	3.361
Modeled PUE	1.551	1.489
PM compliance	72.22%	98.61%
Cooling O&M	EUR 425,432.79	EUR 357,924.84

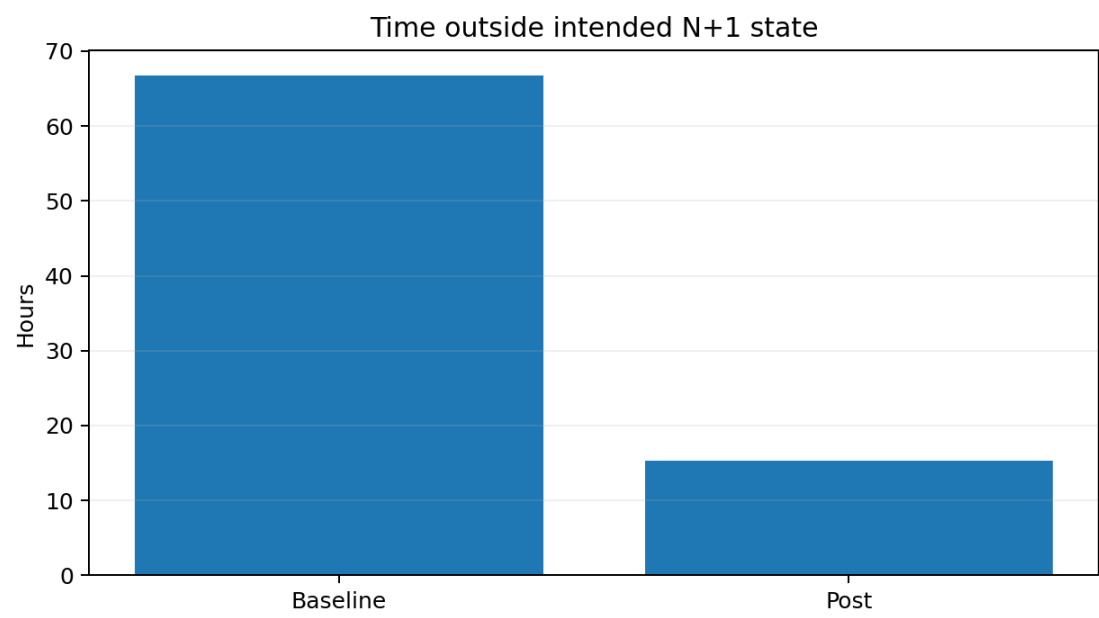
12. Benefit realization

The seeded reference realization produces direct annual cooling O&M savings of EUR 67,507.95, or 15.87%. With EUR 360,000 authorized CAPEX, simple payback based only on direct O&M savings is 5.33 years. Avoided customer outages and Expected Annual Loss reduction are intentionally excluded from the base payback to avoid unsupported monetization or double counting.

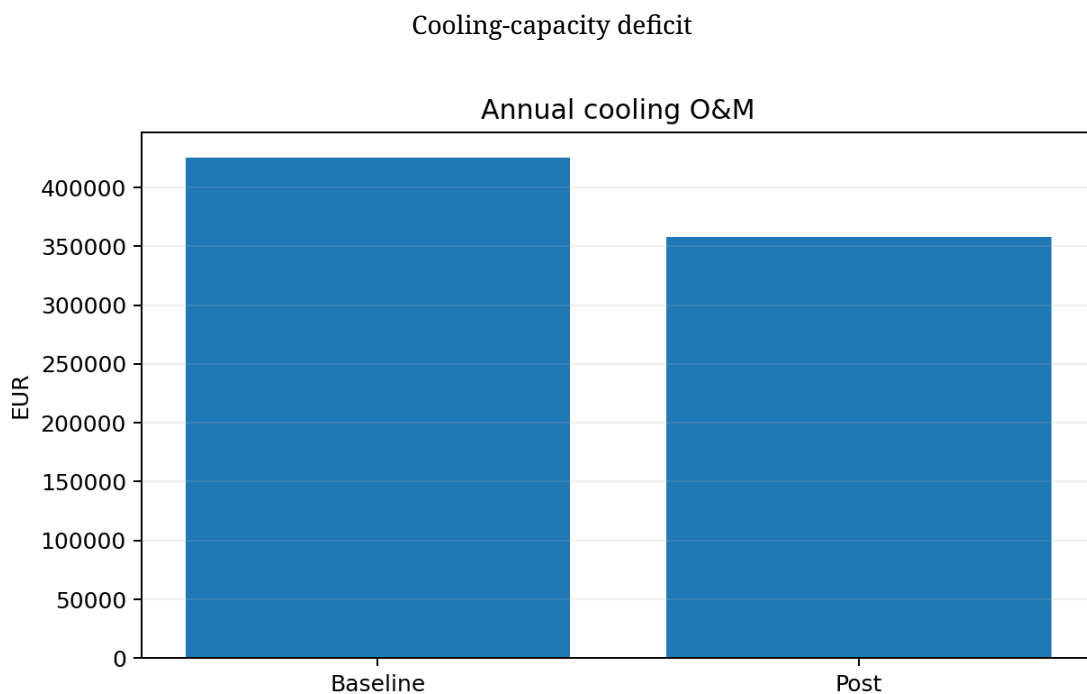
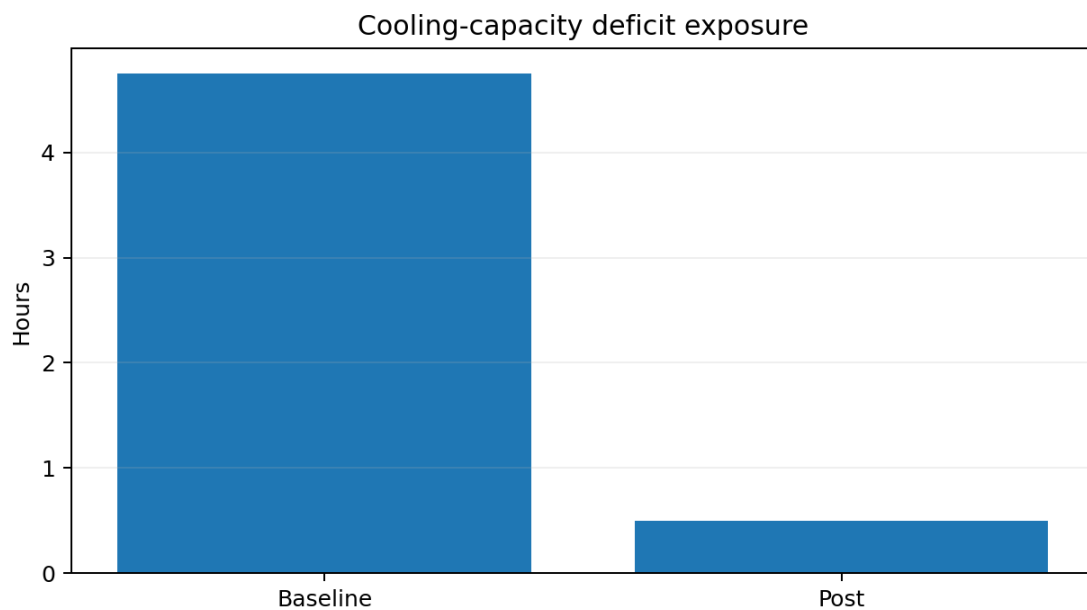
13. Visual evidence



Equipment downtime



Redundancy exposure



Annual cooling O&M

14. Reproducibility and acceptance discipline

Targets are inputs; outcomes are outputs. The model distinguishes a frozen baseline of assumptions, one seeded stochastic reference realization and downstream acceptance results. If evidence contradicts the intended story, the story changes - not the evidence.

The Python v6.1 pipeline has been executed successfully from current evidence: synthetic evidence generation, validation checks, automatic KPI regeneration, the complete pytest suite, requirement verification, benefit calculation, schedule Monte Carlo and chart generation. Weighted COP uses only paired, deduplicated 15-

minute observations where both cooling-energy channels are present. PostgreSQL E2E remains the final release gate before a validated public v1.0 tag.

15. GitHub execution

In GitHub Codespaces: `chmod +x scripts/*.sh`, then run `./scripts/run_python_pipeline.sh`. The pipeline deletes stale derived outputs and regenerates KPI, V&V, benefit, Monte Carlo and chart evidence. For the public release gate, run `./scripts/run_full_validation.sh` with PostgreSQL/psql available. The SQL layer reloads the integrated PM+SE evidence model and independently recalculates mapped KPIs before Python-SQL comparison.

16. Portfolio narrative

This case demonstrates a chronological engineering story: operational deterioration -> evidence -> engineering assessment -> stakeholder needs -> verifiable requirements -> alternatives -> business case -> project authorization -> PM + SE integrated execution -> integration -> verification -> validation -> transition -> post evidence -> benefit realization.

The strongest message is not that every KPI improved. It is that the result is traceable to declared assumptions and reproducible evidence, and that asset-level events are not misrepresented as customer-service outages.

17. Evidence-integrity controls added in v6.1

The v6.1 revision closes the evidence-chain findings identified during independent rerun: weighted COP now uses paired 15-minute energy observations; Python KPI evidence is regenerated on every run; the complete test suite must pass; governed reporting periods are restored without driving service-availability arithmetic; PM+SE entities are restored to PostgreSQL; the legacy telemetry file is removed; and the validation manifest records hashes of generated evidence.

Seeded paired-COP evidence: baseline 2.8087 from 34,744 paired observations (99.155% coverage); post 3.3607 from 34,864 paired observations (99.226% coverage).